AI Essay Detector and Humanizer Application

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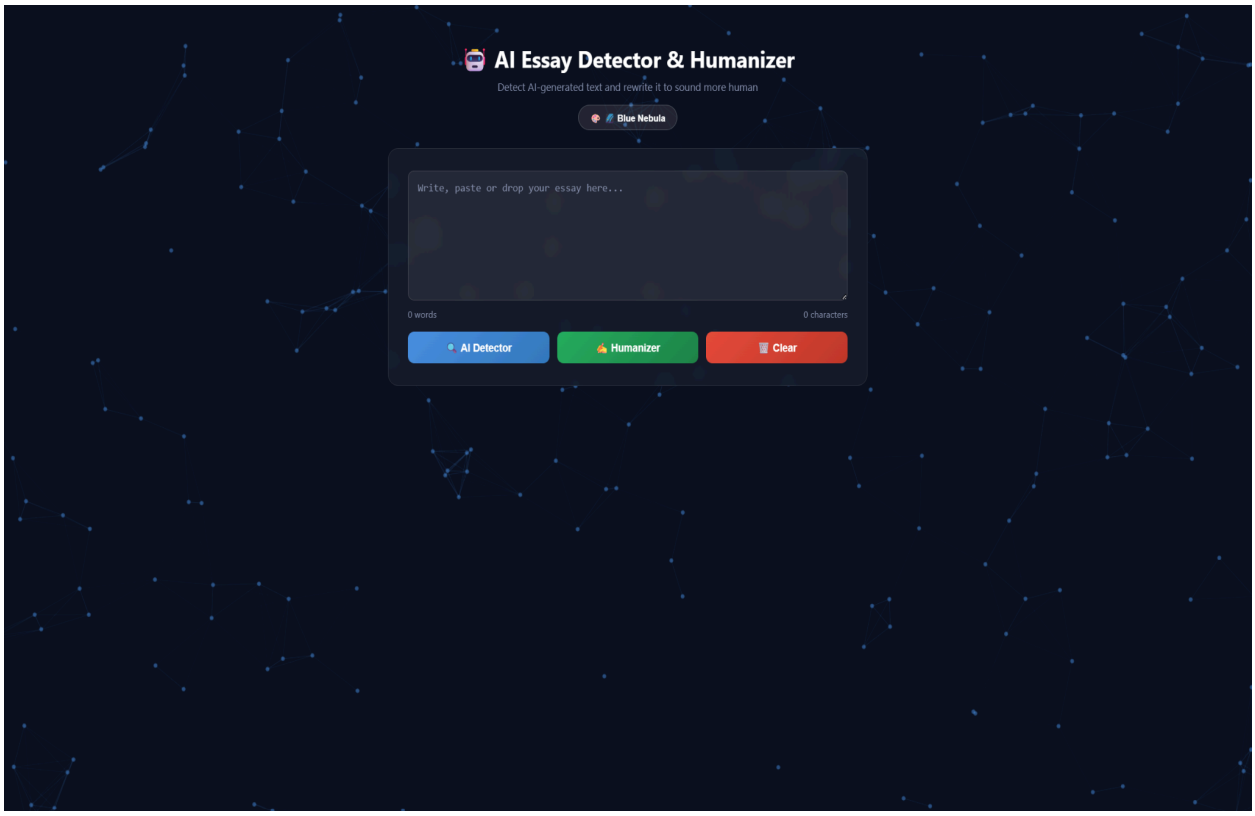

Abstract

With the rapid advancement of AI writing tools, it has become increasingly difficult to distinguish between human-written and AI-generated content in academic writing. This web app brings together AI detection and natural language processing to detect AI-written essays and rephrase them to sound more human-like. It takes in any essay or piece of text and intelligently rewrites the content while maintaining the original meaning in a more natural and human-like writing style. This project aims to address these challenges through a full-stack web application. The application was found to be fairly competent at choosing an AI essay from one written by a human.

Introduction

The use of AI has become significantly more prevalent in academic writing, whether that be essays, academic papers, or even thesis papers. While some of it feels homogenous, like it was written by a robot, it's not always the easiest to pick it out from something that was simply well written by a person with an extensive vocabulary. However, there are patterns that tend to emerge from extensive use of AI, especially in longer papers and academic works. In addition, sometimes it can be difficult to understand a lot of big words that people like to use in academic papers and essays, words that someone may never have seen before. AI has a tendency to use many of these words when writing academic content. Many times it feels inhuman, almost like it wasn't written by a human, typically because it wasn't. For this reason, we created an application that both detects AI usage in academic writing as well as takes that AI generated writing and turns it into a more "natural" format that anyone should be able to understand. This was done by using the Sapling AI Detection API to find the percentage of a given essay/academic work that was likely written by AI, then having that essay run through Groq LLaMA 3.3 70B model with the purpose of creating a more "human" sounding version.

Related Work

We used several tools and research efforts to explore about AI text detection and Humanization:

For text detection: The GPTZero tool , which relies on measuring burstiness and perplexity, was one of the first free online tools capable of identifying whether a text was written by humans or produced through AI. Analogously, Sapling AI offers an online detection API which calculates the probability of AI-based writing.

Regarding humanization, there has been research in paraphrasing techniques with models like T5 and PEGASUS for rewriting text while retaining its meaning. The newer models like GPT-4 and LLaMA have also proven their abilities in style transfer and text rewriting.

Our project stands apart from other existing works in multiple aspects. Our approach is not confined to detection or rewriting but rather encompasses both processes in an integrated web application. Also, our work provides a quantifiable measure of the text alteration post humanization through cosine similarity, which is not available in other existing works.

Data

We used a dataset named “Human vs AI Generated Essays” Dataset by Navjot Kaushal (Kaggle) as it provides dataset which includes essays categorized under AI-generated essays (labels=1) and human-written essays (labels=0). The text field holds complete paragraphs on diverse subjects such as technology, education, and science. The complete dataset comprises hundreds of labeled data samples. For the purpose of evaluating the model, 100 essays have been selected using stratified sampling based on the two categories (with fixed random seed for consistency). Very minimal preprocessing had to be performed for the data. Each paragraph of each essay was restricted to 2,000 characters as required by the API constraints. Labels were also converted from numeric labels to string labels (Human Written / AI Generated).

Methods

- System Architecture: Our application follows a standard client-server architecture
  Frontend: HTML, CSS, JavaScript with an animated particle background
  Backend: Python Flask handling two routes, detection and humanization
  APIs: Sapling AI for detection, Groq LLaMA 3.3 70B for humanization

- AI Detection: For our detection module sends user-submitted text to the Sapling AI Detection API via an HTTP POST request. The API returns a score between 0 and 1, which is converted to a percentage. Scores above 70% are labeled AI Generated, scores between 40–70% are flagged as mixed, and scores below 40% are labeled Human Written.

- Text Humanization: The humanization module sends the essay to the Groq API using the LLaMA 3.3 70B model. A carefully engineered prompt instructs the model to rewrite the text in a natural, conversational tone while preserving the original meaning. Key prompt constraints include avoiding formal vocabulary, using contractions naturally, and not adding new information.

- Rewriting Analysis: To measure how much the text changed after humanization, we compute the cosine similarity between the original and rewritten text using scikit-learn. The Words Changed percentage is calculated as: Words Changed % = (1 - cosine_similarity) × 100.

Experiments & Results

For experimenting purposes we used a dataset named “Human vs AI Generated Essays” from kaggle for which our system showed 90 percent accuracy on the test dataset of 100 samples. Misclassifications, if any, were found to be false negatives, with the system identifying a sample text written by AI as being written by a human. False positive results were not observed.

Text	Original Score	Words Changed
Sample 1	100 %	72%
Sample 2	90%	70%
Sample 3	80%	65%

The above table shows approximation about the samples we tested

Future Work: Integrate a more powerful detection model trained on paraphrased content, Add sentence-level highlighting to show which parts of the text are most AI-like, Support file uploads for .txt and .pdf documents, Deploy the application online for broader accessibility and Expand the evaluation dataset to 500+ essays for more statistically significant results.

Conclusion

For this project we successfully developed a web application that combines AI text detection and humanization into a single interactive tool. The detection module was able to reach 90% accuracy when run on an unlabeled dataset from the real world. The humanization module, based on the Groq LLaMA 3.3 70B model, was capable of rewriting texts in a more natural way with the number of modified words ranging between 65-75% per test sample.

Limitation: The primary drawback is that the Sapling detector always detected AI-generated text as such, which indicates that existing humanization methods are inadequate for fooling advanced detectors. Another limitation is that there are rate restrictions when using Sapling’s free version.

Citations/References

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